MIL-T-15089C (OS)
13 September 1974
SUPERSEDING
MIL-T-15089B (NOrd)
27 April 1959

PROPOSED
MILITARY SPECIFICATION

TUBING, ALUMINUM ALLOY, ROUND, SEAMLESS (FOR ROCKET MOTORS)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers aluminum alloy, seamless, round tubing for use in the manufacture of rocket motors.

*1.2 **Classification.** The tubing shall be of the following alloys, tempers, and classes as specified (see 6.1):

Alloy 2014 - T6 or T6511
Alloy 2024 - T81 or T8511

Class	Nominal size	
	Outside diameter (inches)	Wall thickness (inches)
1	5.25	0.292
2	5.00	0.138
3	2.75	0.072
4	2.00	0.060

FSC 4710

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bid or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-H-6088	Heat Treatment of Aluminum Alloys; Process for (for Aircraft Applications)

STANDARDS

Federal

*FED-STD-151	Metal, Test Methods
*FED-STD-245	Tolerance for Aluminum Alloy and Magnesium Alloy Wrought Products

Military

MIL-STD-649	Aluminum and Magnesium Products, Preparation for Shipment and Storage

DRAWINGS

Navy Department

If required by the contract or order, applicable drawings for the finished rocket motor form a part of this specification (see 6.1).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials

ASTM E 8 — Standard Method of Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 **Chemical composition.** The chemical composition shall conform to the requirements given in table I.

*Table I

CHEMICAL COMPOSITION

Element	Alloy 2014		Alloy 2024	
	(%, min.)	(%, max.)	(%, min.)	(%, max.)
Copper	3.9	5.0	3.8	4.9
Magnesium	0.20	0.8	1.2	1.8
Manganese	0.40	1.2	0.30	0.9
Silicon	0.50	1.2	—	0.50
Iron	—	0.7	—	0.50
Chromium	—	0.10	—	0.10
Zinc	—	0.25	—	0.25
Titanium	—	0.15	—	0.10
Titanium plus zirconium	—	0.20	—	0.25
Others (each)[1]	—	0.05	—	0.05
Others (total)[1]	—	0.15	—	0.15
Aluminum	Remainder		Remainder	

[1]Analysis shall regularly be made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that these other elements are not present in excess of 0.05 percent each, and that the total of these other elements does not exceed 0.15 percent. The material shall not contain these other elements in excess of the above limits.

3.2 **Heat treatment.**

3.2.1 **Process.** The process for heat treatment shall be in accordance with the requirements of MIL-H-6088, except as modified herein.

3.2.2 **Corrosion.** Tubing shall not show presence of excessive intergranular attack when tested in accordance with 4.5.2.

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3.3 **Mechanical properties.** Tubing shall conform to the minimum requirements shown in table II.

*Table II

MECHANICAL PROPERTIES

Alloy and temper	Class	Tensile strength (psi, min)	Yield strength at 0.2% offset (psi, min)	Elongation in 2 inches (%, min)
2014-T651, T6511	All	60,000	53,000	7
2024-T81, T8511	1	66,000	58,000	5
	2, 3, 4	64,000	56,000	4

*3.4 **Internal hydrostatic pressure.** When specified, tubing shall be capable of conforming to the hydrostatic pressure test requirements specified on the applicable drawings for the finished rocket motor (see 6.1).

*3.5 **Dimensional tolerances.** Unless otherwise specified, tolerances shall conform to FED-STD-245.

3.6 **Marking.** Each length of tubing shall be legibly marked parallel to the axis of the tubing at intervals not to exceed nominal length for the finished rocket motor tube (see 6.1). Marking fluid shall be such that the markings produced shall not be removable by normal handling. Marking shall include the following:

a. Manufacturer's name or trademark
b. Number of this specification
c. Class number, alloy, and temper designation
d. Lot number tubing

*3.7 **Workmanship.** The tubing shall be seamless and uniform in quality and temper. The exterior and interior surfaces shall be clean, smooth, and free from slivers, laminations, folds, grooves, cracks, and other injurious defects within the limits consistent with the best commercial practice. Discoloration due to thermal treatment will not be cause for rejection.

*4. QUALITY ASSURANCE PROVISIONS

*4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use

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his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 **Sampling.**

*4.2.1 **Lot.** A lot shall consist of an identifiable quantity of tubing of the same alloy, temper, and class, traceable to a heat treat lot or lots, and subjected to inspection at one time.

*4.2.2 **Sampling for chemical analysis.**

*4.2.2.1 **Ingot analysis.** At least one sample shall be taken from each group of ingots of the same alloy poured simultaneously from the same source of molten metal by the producer and analyzed to determine conformance to 3.1. Ingots not conforming to the requirements of this specification shall be rejected. Complete ingot analysis records shall be available, at the producer's plant, to the procuring activity.

*4.2.2.2 **Finished product analysis.** When compliance with 4.2.2.1 cannot be established, samples shall be selected as follows: From tubing having a nominal weight of less than 1 pound per lineal foot, one sample shall be selected from each lot weighing 1,000 pounds or less; from lots weighing more than 1,000 pounds, one additional sample shall be taken for each 1,000 pounds or fraction thereof in excess of the first 1,000 pounds. From tubing having a nominal weight of 1 pound or more per lineal foot, one sample shall be taken from each lot consisting of 1,000 feet or less; from lot consisting of more than 1,000 feet, one additional sample shall be taken from each 1,000 feet or fraction thereof in excess of the first 1,000 feet. Only one test specimen shall be taken from any one piece when more than one piece is available. Not more than one analysis shall be required per piece.

*4.2.3 **Samples for tests.**

*4.2.3.1 **Samples for tensile tests.**

*4.2.3.1.1 **Number of test samples in temper supplied.** From tubing having a nominal weight of less than 1 pound per lineal foot, one tension-test sample shall be selected from each lot weighing 1,000 pounds or less; from lots weighing more than 1,000 pounds, one additional sample shall be taken for each 1,000 pounds or fraction thereof in excess of the first 1,000 pounds. From tubing having a nominal weight of 1 pound or more per lineal foot, one tension-test sample shall be taken from each lot consisting of 1,000 feet or less;

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from lots consisting of more than 1,000 feet, one additional sample shall be taken for each 1,000 feet or fraction thereof in excess of the first 1,000 feet. Only one tension test specimen shall be taken from any one piece when more than one piece is available.

*4.2.4 **Intergranular corrosion.** At least two samples shall be selected from each lot of tubing for the intergranular corrosion tests. Samples shall be taken from different lengths of the tubing and shall each include the complete cross-section of the tubing. Samples shall be taken from the end of the tubing which enters the quenching bath last. At least four cross-section specimens, located 90° apart, shall be cut from each sample for microscopic examination in accordance with FED-STD-151, method 822.

*4.2.5 **Sampling for visual and dimensional examination.** All tubing shall be examined to determine conformance to this specification with respect to workmanship and identification marking. Examinations for dimensions shall be made to insure conformance to the tolerances specified. On approval of the procuring activity, the supplier may use a system of statistical quality control for dimensional, marking, and workmanship examination.

*4.3 **Examination.**

*4.3.1 **Visual and dimensional examination.** Each sample tube selected in accordance with 4.2.4 shall be visually examined and measured to verify conformance to this specification.

*4.3.2 **Preservation, packaging, packing, and marking for shipment.** The preservation, packaging, packing, and marking for shipment shall be examined to determine compliance with the requirements of section 5.

*4.4 **Test methods.**

*4.4.1 **Chemical analysis.** Chemical analysis shall be made by the wet chemical method in accordance with FED-STD-151, method 111, or by the spectrochemical method in accordance with FED-STD-151, method 112.

*4.4.2 **Mechanical properties.**

*4.4.2.1 **Types of specimens.** Specimens for tensile testing shall conform to the requirements of ASTM E 8. When practicable, tubing should be tested in full section. For tubing which is not tested in full section, the standard 1/2-inch-wide longitudinal tension test specimen (specimen No. 1 in figure 13 of ASTM E 8) for large diameter tubular products shall be used.

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*4.4.2.2 **Location of test specimens.** Tension test specimens shall be taken from the tubing in the longitudinal direction.

*4.4.2.3 **Tensile tests.**

*4.4.2.3.1 **Tensile strength.** Tensile strength shall be determined in accordance with ASTM E 8.

*4.4.2.3.2 **Yield strength.** The yield strength shall be determined by the offset method in ASTM E 8.

*4.4.3 **Intergranular corrosion.** Intergranular corrosion susceptibility shall be determined in accordance with FED-STD-151, method 822.

*4.5 **Acceptance and rejection criteria.** If any specimen fails to conform to the requirements of this specification, it shall be cause for rejection of the material represented by the specimen subject to the retest provisions of FED-STD-151. When no sampling plan is approved by the procuring agency, and when there is evidence that indicates that a failed specimen was not representative of the lot of material, at least two specimens shall be selected to replace each test specimen which failed. All specimens so selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

5. PREPARATION FOR DELIVERY

*5.1 **Preservation, packaging, and packing.** All tubing shall be preserved, packaged, and packed in accordance with the requirements of MIL-STD-649. Unless otherwise specified (see 6.2), preservation, packaging, and packing shall be level C.

*5.2 **Marking.** In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with MIL-STD-649.

6. NOTES

*6.1 **Ordering data.** Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Alloy, temper, and class (see 1.2)

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c. Applicable drawings (see 2.1)

d. Internal hydrostatic pressure testing, if required (see 3.4)

e. Tolerances other than those specified in FED-STD-245, if required (see 3.5)

f. Level of preservation, packaging, and packing (see 5.1)

g. Special marking, if required (see 5.2)

*6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-15089C (OS), Tubing, Aluminum Alloy, Round, Seamless (For Rocket Motors)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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